

Shanxiao Wang, PhD

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TECHNICAL SKILLS

Modeling	Econometric modeling, statistical modeling, macroeconomic and DSGE modeling, time-series modeling, linear and logistic regression, supervised and unsupervised machine learning, survival and hazard models, simulation and optimization methods, causal inference methods, model diagnostics and evaluation; LLM tools and workflows, NLP applications and deployment
Languages	Python, SQL, R, MATLAB, Stata, SAS
Data	Data processing, data management, data visualization, and dashboard development

PROJECTS

Credit Stress-Testing Engine

- Developed an end-to-end consumer credit loss-forecasting pipeline using more than 2 million LendingClub loan records.
- Built three modeling components: a leakage-safe probability-of-default (PD) model built with logistic regression and validated out-of-time; a macro satellite model linking portfolio defaults to a robust, correctly-signed macroeconomic channel stable across specifications; and a scenario engine transmitting 2001 and 2008–09 stress paths into expected loss forecasts.
- Diagnosed and corrected a survivorship-bias flaw in an earlier design via a full model rebuild, with documentation of methodology and results.

Macro-Finance PhD Dissertation

- Built a DSGE model with heterogeneous retail and wholesale banks that differ in leverage constraints and balance-sheet behavior. Showed how the expansion of wholesale banking amplifies the transmission of financial risk shocks into greater business-cycle volatility; calibrated to U.S. Flow of Funds data.
- Incorporated financial regulation as a structural model parameter, quantifying how interbank deregulation drove the post-1980s growth of wholesale banking.
- Estimated a structural VAR on quarterly U.S. macro data (1986–2019) with Cholesky identification and Monte Carlo confidence bands, quantifying how macroeconomic uncertainty transmits through bank leverage and balance sheets to aggregate investment and output.

Booster Landing Prediction

- Built an end-to-end machine-learning pipeline to predict rocket-booster landing outcomes, including data collection through APIs and web scraping, data cleaning, feature engineering, exploratory data analysis (EDA) using SQL and visualization, model development, and validation.
- Built and compared classification models (logistic regression, SVM, decision tree, KNN) to predict first-stage landing success from launch features. Evaluated model performance using test-set accuracy and confusion matrices, with 10-fold cross-validated grid search for hyperparameter tuning.
- Deployed a live dashboard and an interactive map analyzing launch site geography.

Additional projects:

- shanxiaow.github.io
- github.com/shanxiaow

EDUCATION & CERTIFICATION

PhD, Economics (Macro-Finance)

2021

University of Maryland, College Park

- Dissertation: *Essays on Wholesale Banking and the Macro-Economy* — DSGE and structural VAR analysis of risk shocks, bank leverage, and economic outcomes.
- Graduate training in financial economics, econometrics, numerical and computational methods, including optimization, Monte Carlo simulation, and Bayesian estimation.

MS, Epidemiology

2009

Johns Hopkins Bloomberg School of Public Health

- Graduate training in statistical analysis and causal inference, including survival analysis and time-to-event (hazard) models.
- Advanced training in research methodology. Completed thesis-based graduate research using a large longitudinal dataset.

Bachelor of Science, Peking University

2007

IBM Data Science Professional Certificate

2023

EXPERIENCE

Faculty, School of Data Science; School of Economics

2022 – 2025

Yokohama City University — Yokohama, Japan

- Designed and delivered courses in intermediate economics, statistical methods, quantitative modeling, and R and Python programming, with sole responsibility for curriculum design, instruction, and evaluation.
- Translated complex quantitative methods into clear lectures, assignments, and feedback for students with varying technical backgrounds.

Instructor, Department of Economics

2014 – 2017

Teaching Assistant, MS in Applied Economics Program

2017 – 2019

University of Maryland, College Park — College Park, MD

- Taught courses in statistical analysis and computational methods in economics.
- Supported graduate coursework in macroeconomics, microeconomics, econometrics, and financial economics.
- Instructed students in Excel, Stata, MATLAB, and Python for applied economic analysis.

Data Analyst, Demographic and Health Surveys (DHS), USAID

2009 – 2013

ICF International — Calverton, MD

- Managed and analyzed large-scale, nationally representative survey data covering 100+ countries.
- Prepared statistical analyses, outputs, and visualizations for research publications, presentations, and policy briefs.
- Contributed to capacity-building and project evaluation efforts.